



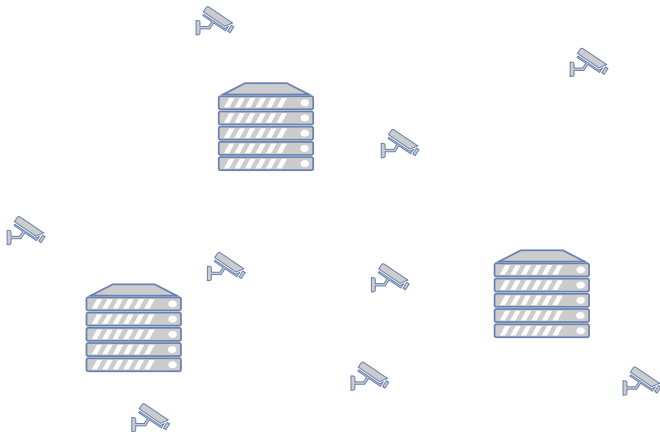
Technische
Universität
Braunschweig



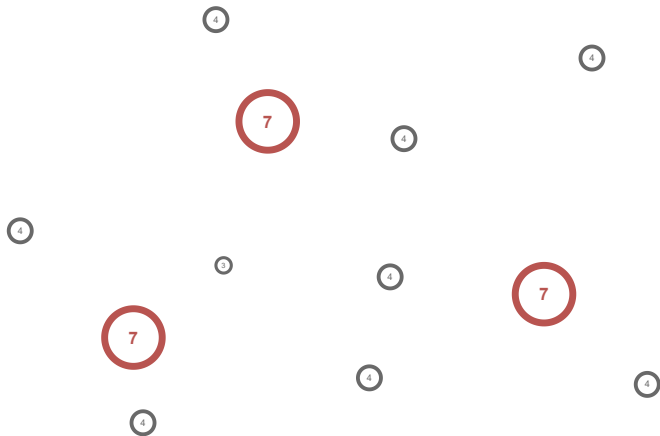
Algorithmische Methoden zu gradbeschränkten Triangulierungen

Fabian Alich, 8. September 2025

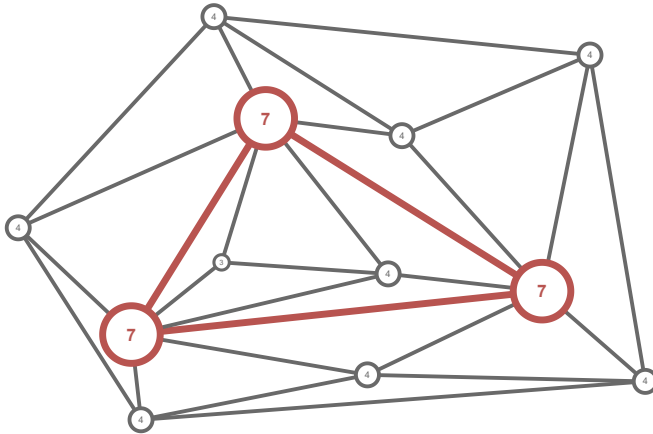
Praktische Relevanz



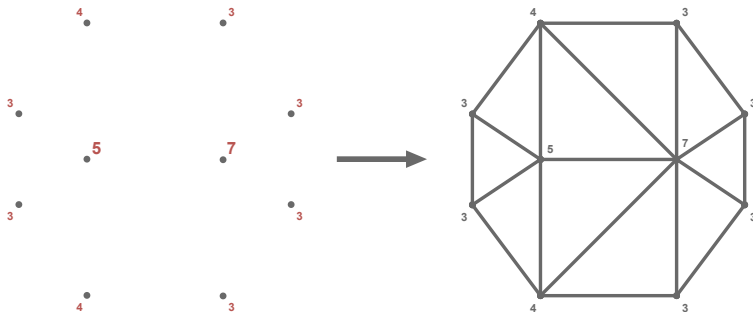
Praktische Relevanz



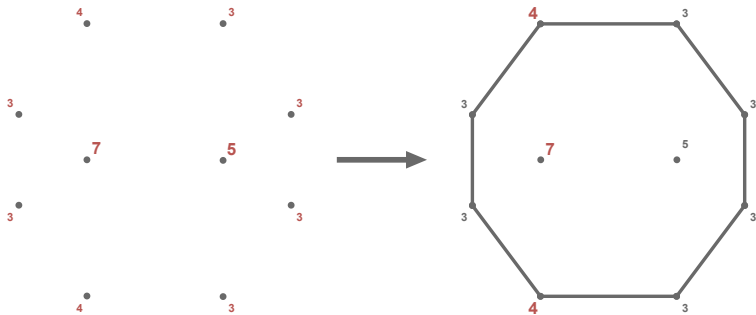
Praktische Relevanz



Ja-Instanz



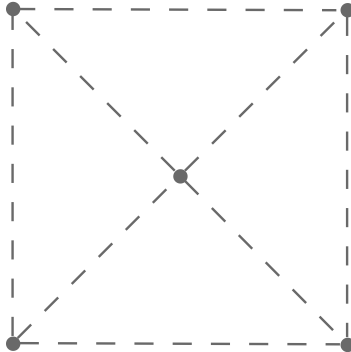
Nein-Instanz



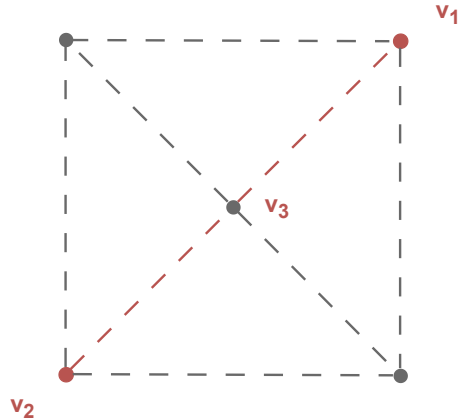
Teil I

Lösungsansätze

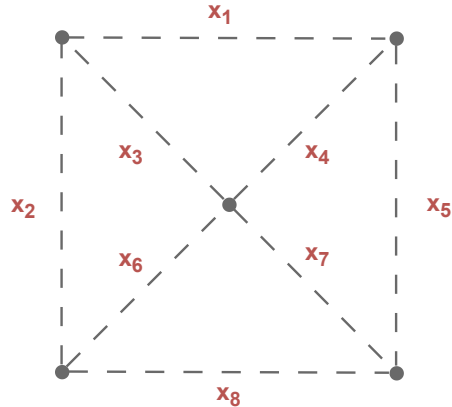
Kantenansatz



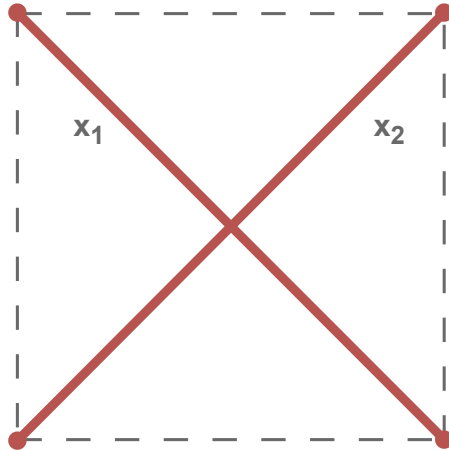
Kantenansatz



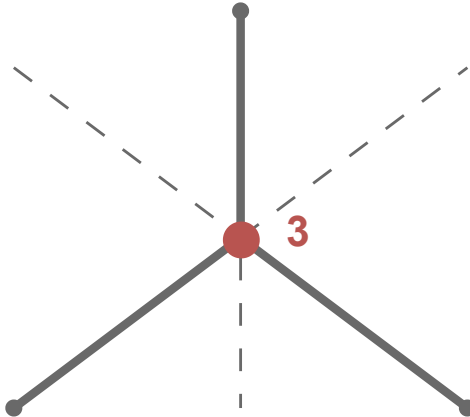
Kantenansatz



Überschneidungsbedingung



Gradbeschränkung (Entscheidungsproblem)

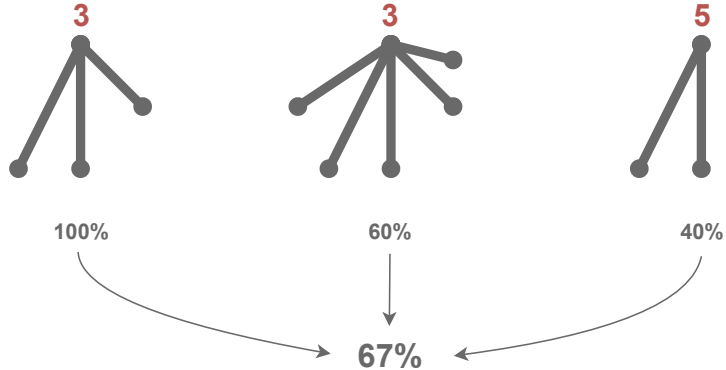


Maximale Kanten (Entscheidungsproblem)

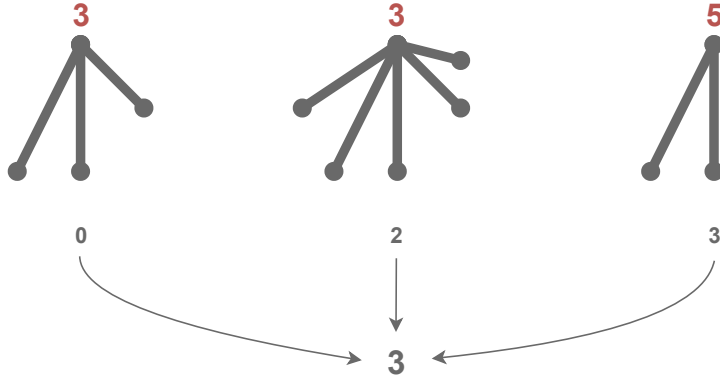
FA: Formal raus?

$$\text{Anzahl Kanten} \cdot 2 \stackrel{!}{=} \sum \text{Sollgrad}$$

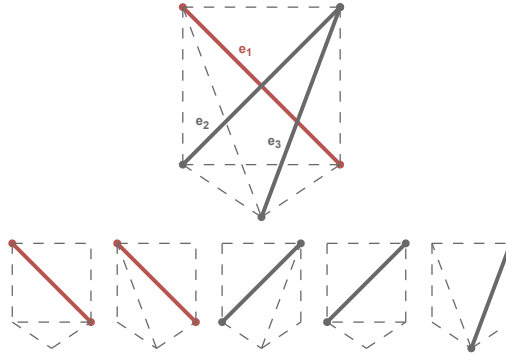
Gradbeschränkung (Optimierungsproblem)



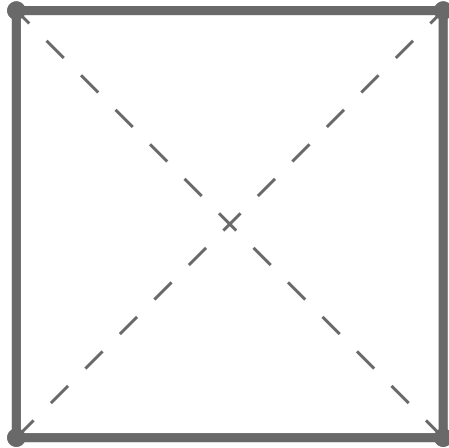
Gradbeschränkung (Optimierungsproblem)



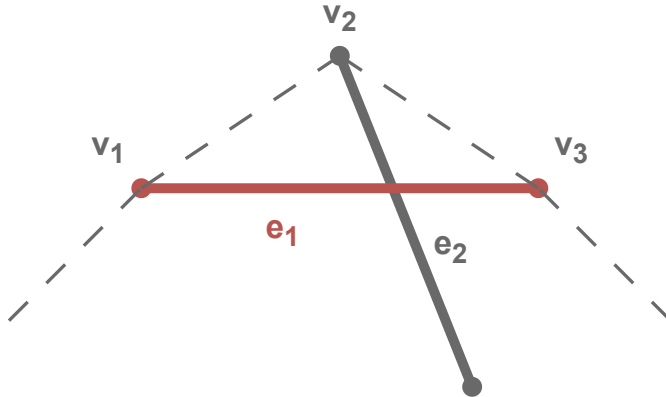
Alternative Kantenbedingung



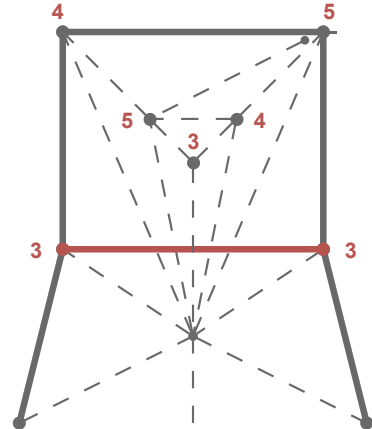
Hülle Fixieren



Kanten Fixieren / Ausschließen



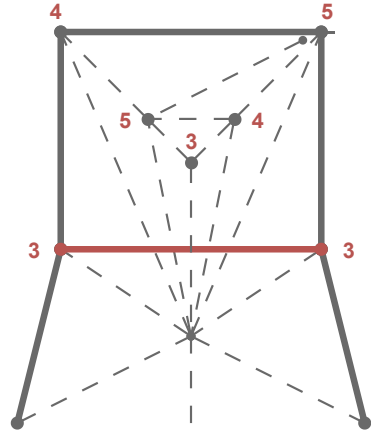
Kanten Ausschließen



Kanten Ausschließen

Anzahl Kanten

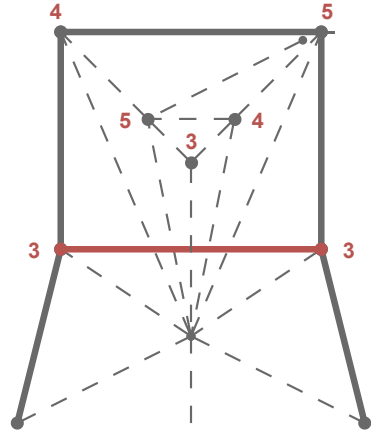
14



Kanten Ausschließen

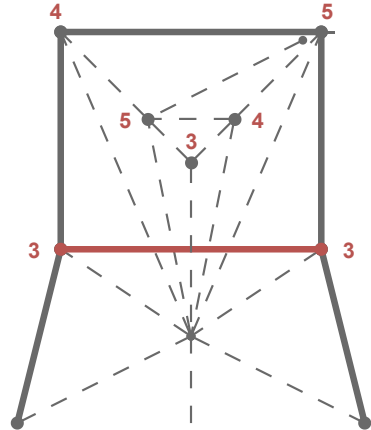
Anzahl Kanten 14

Benötigte Sollgrade 28

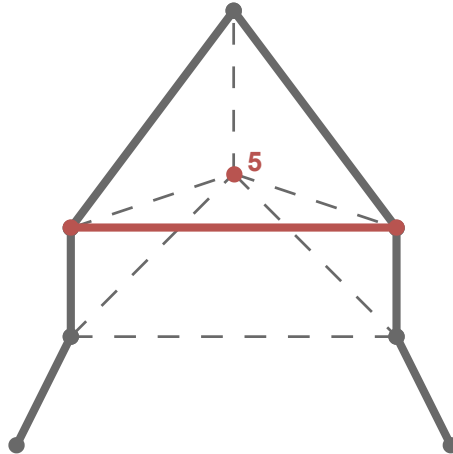


Kanten Ausschließen

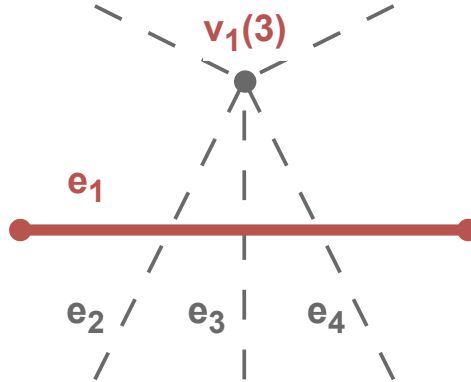
Anzahl Kanten	14
Benötigte Sollgrade	28
Summe Sollgrade	27



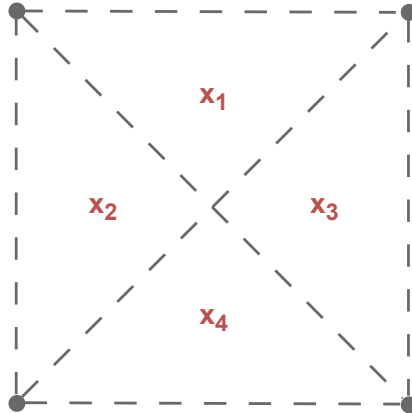
Kanten Ausschließen



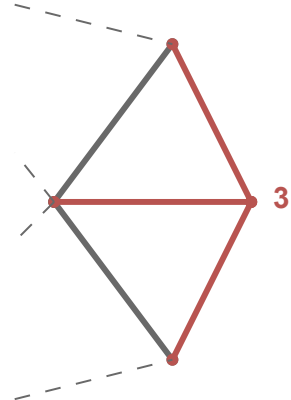
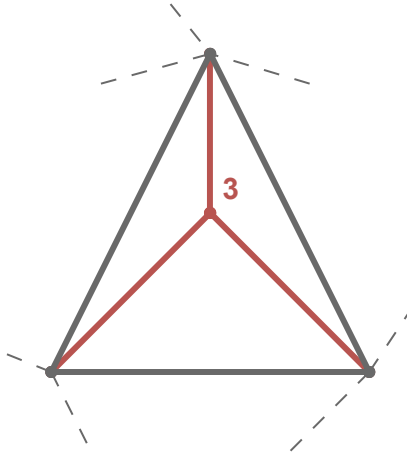
Kanten Ausschließen



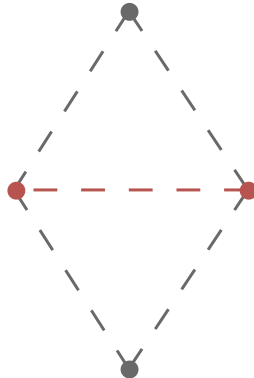
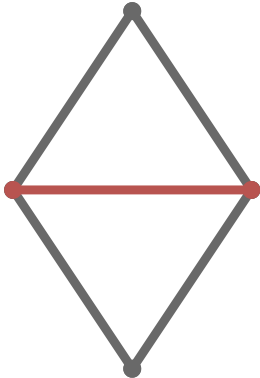
Dreiecksansatz



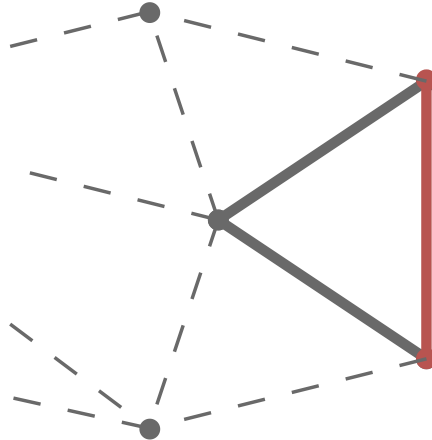
Gradbeschränkung



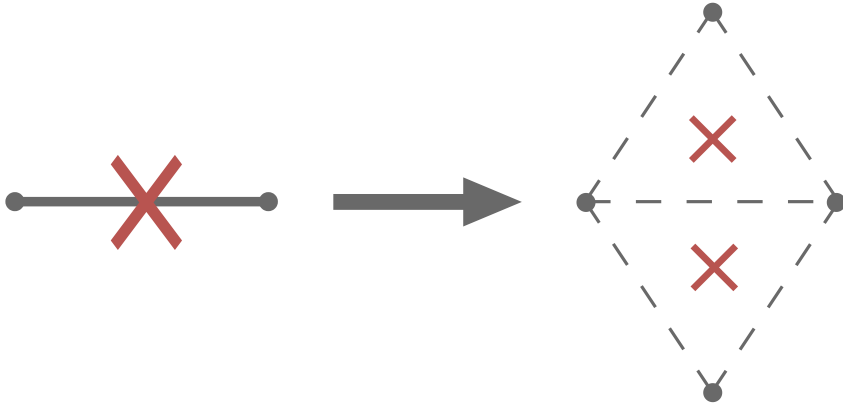
Überschneidungsbedingung



Überschneidungsbedingung



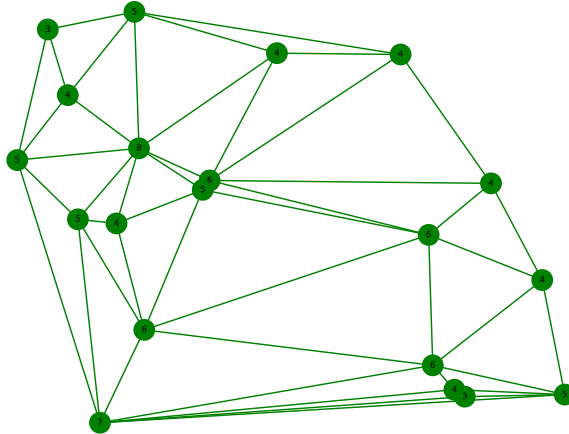
Dreiecke ausschließen



Teil II

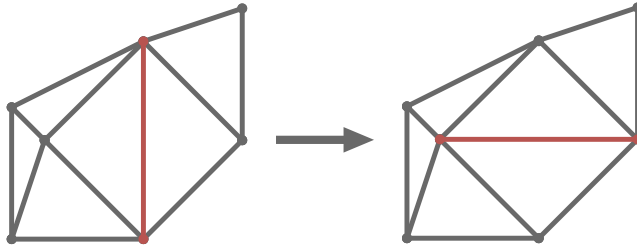
Instanzgenerierung

Delaunay

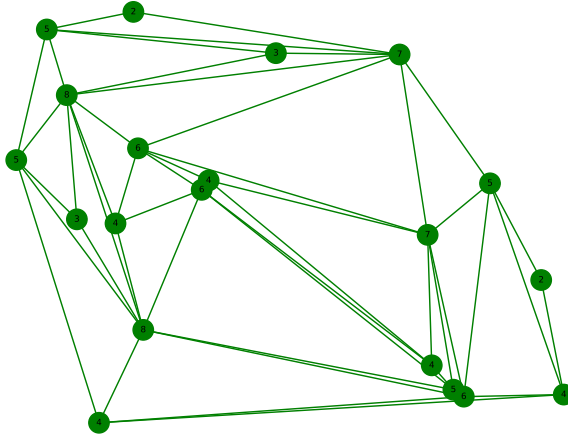


Flip

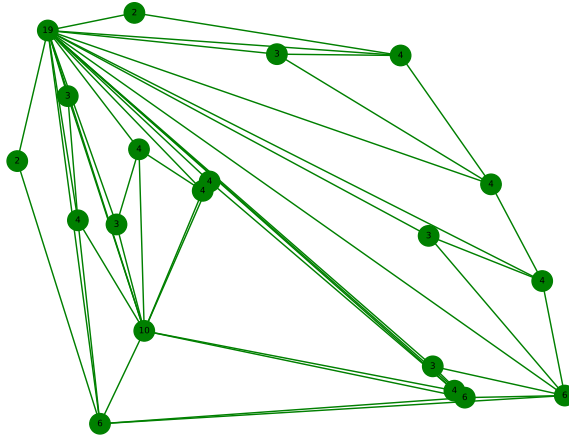
FA: Soll ich das als Animation



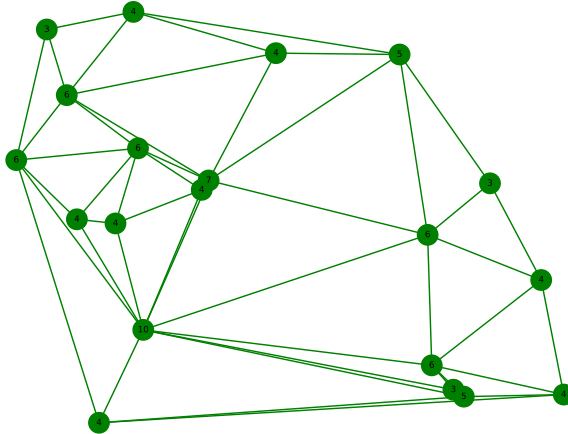
Delaunay-Flips



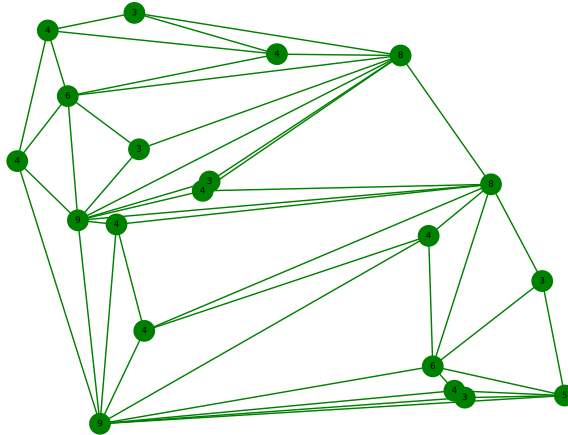
Iterativ



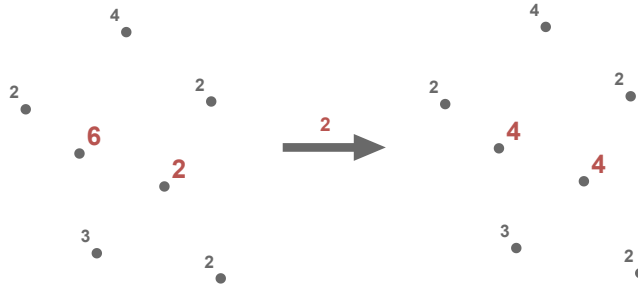
Greedy



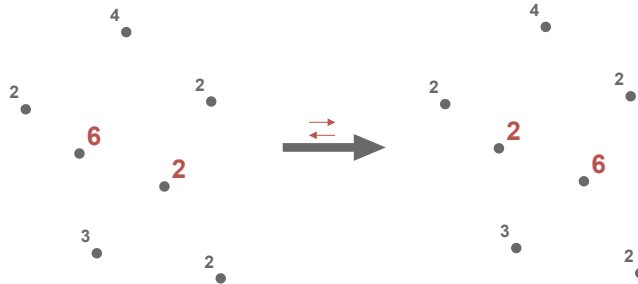
Random



Move



Change



Teil III

Auswertung

Solver



Gurobi

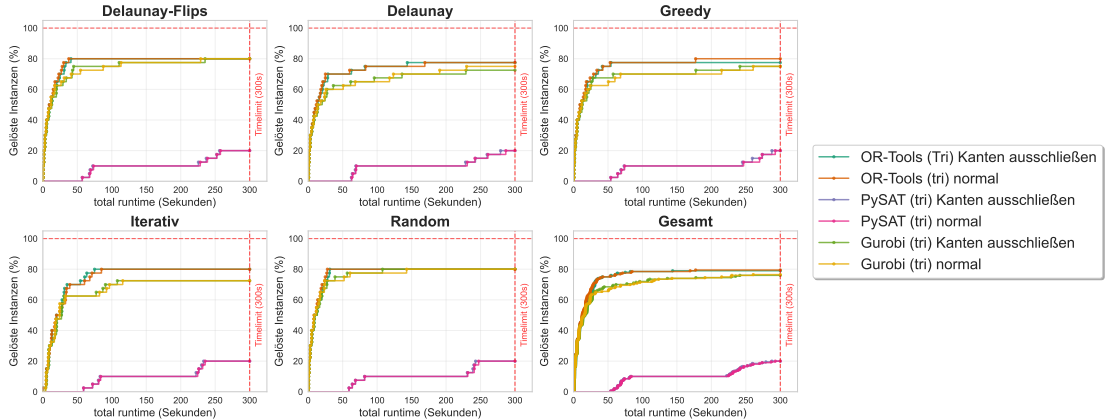


OR-Tools

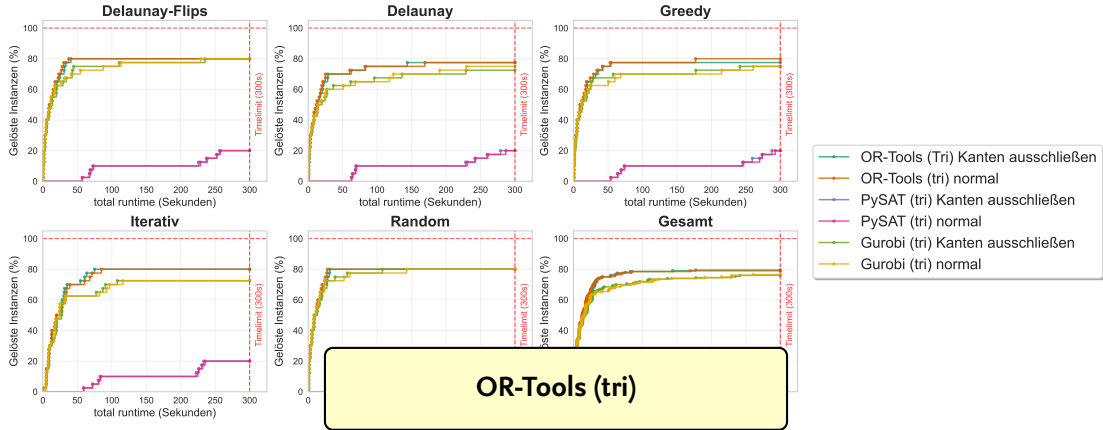


PySAT

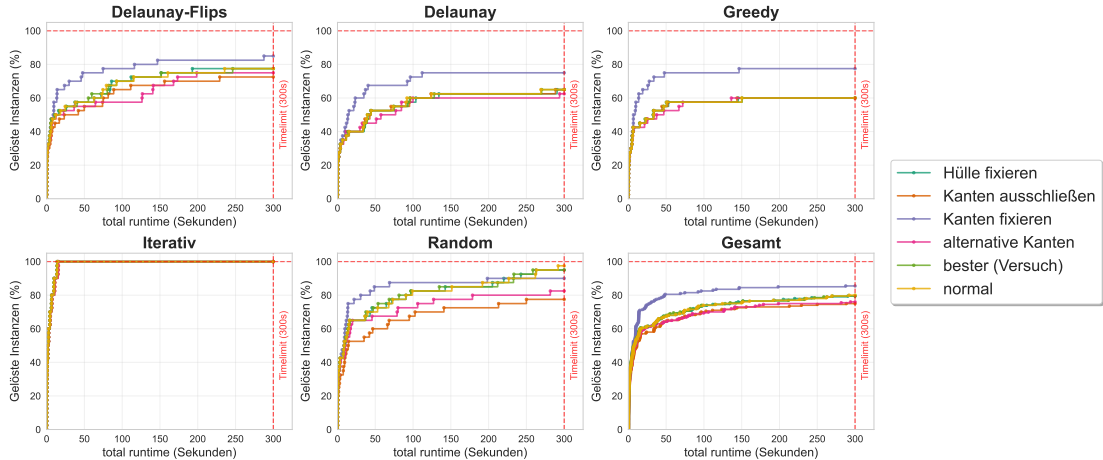
Dreiecke Bedingungen



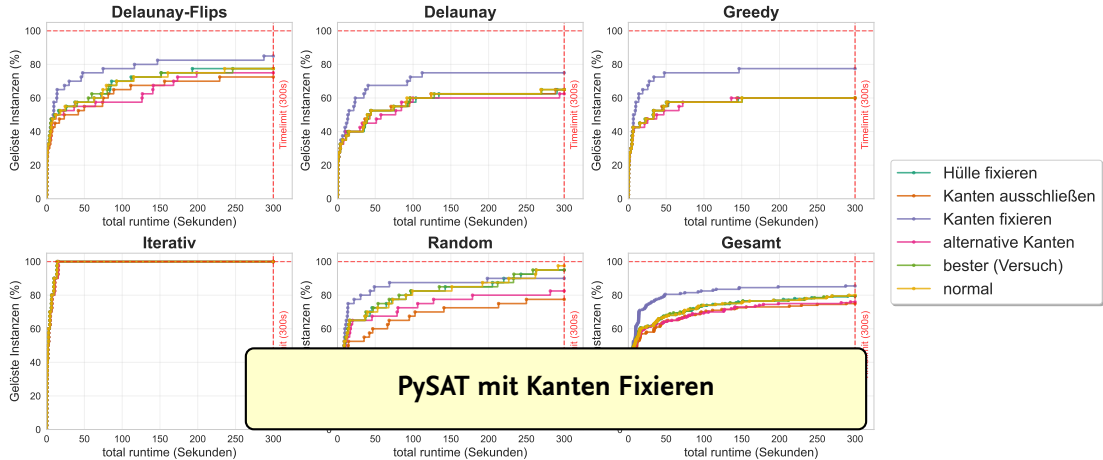
Dreiecke Bedingungen



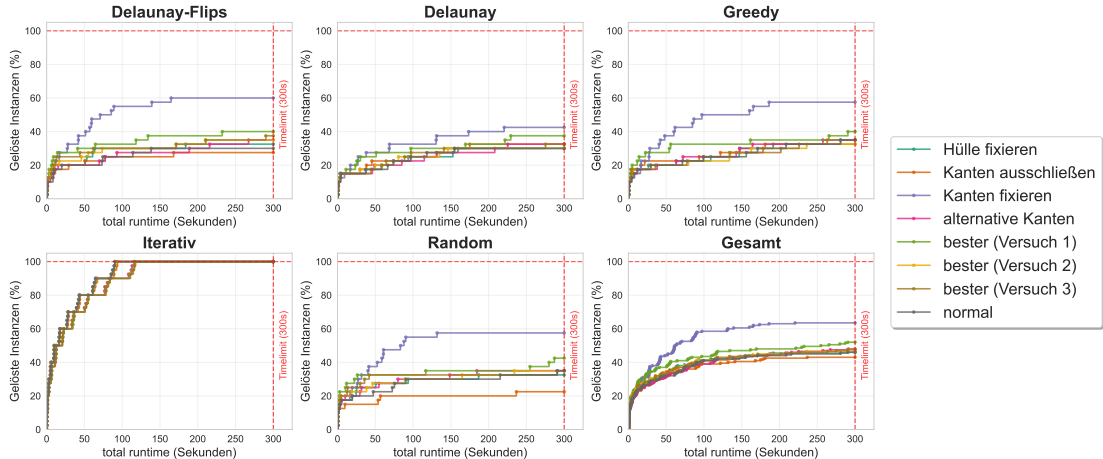
PySAT Bedingungen



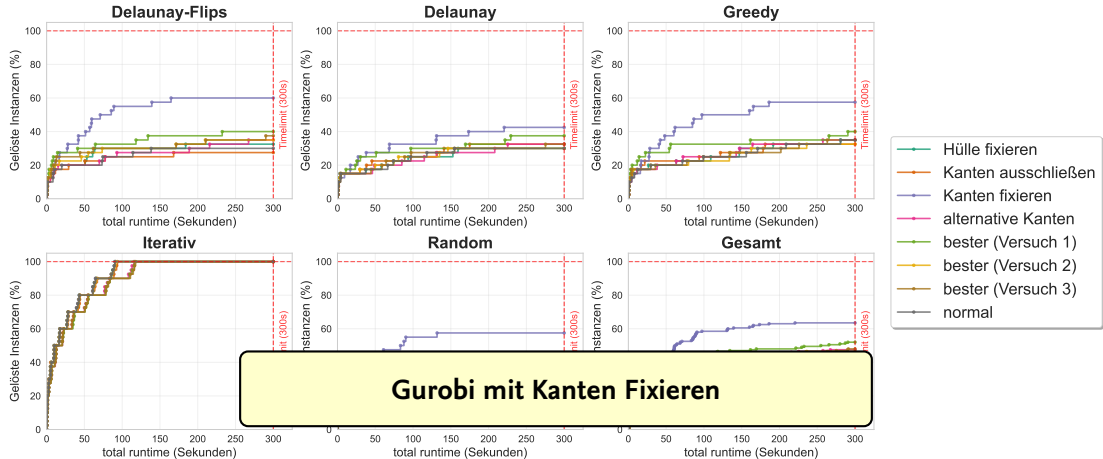
PySAT Bedingungen



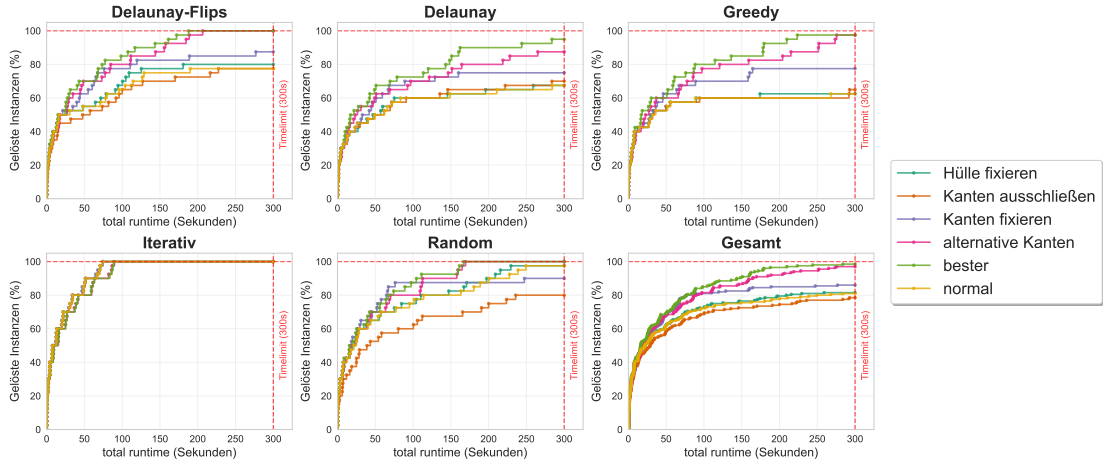
Gurobi Bedingungen



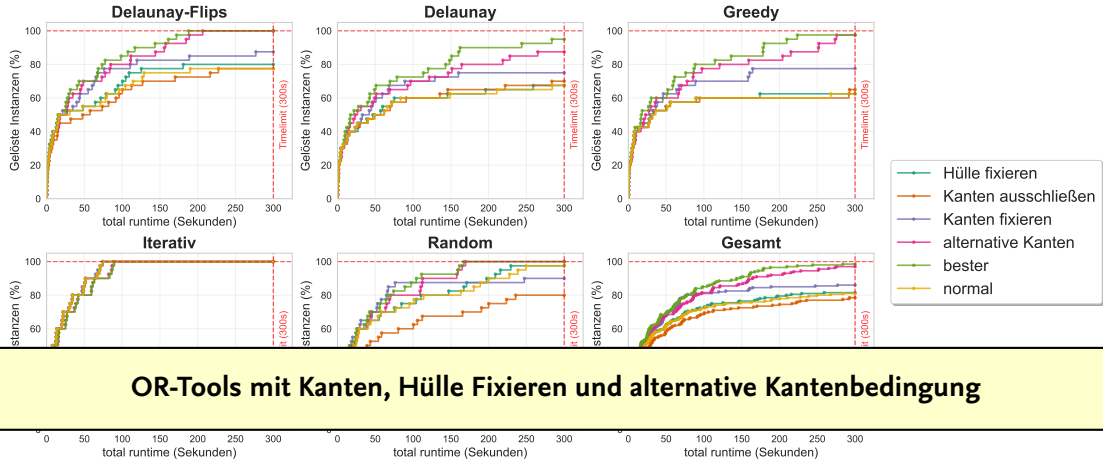
Gurobi Bedingungen



OR-Tools Bedingungen

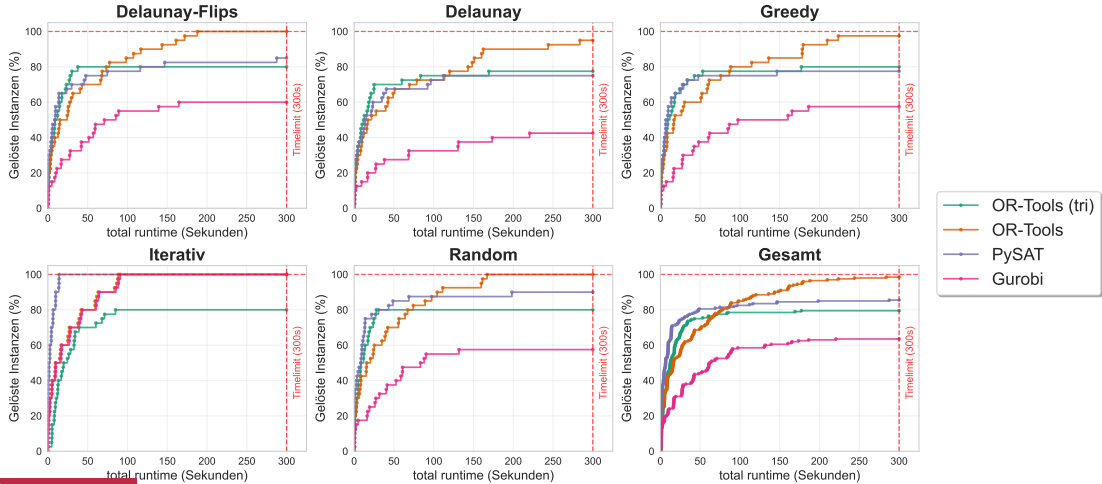


OR-Tools Bedingungen

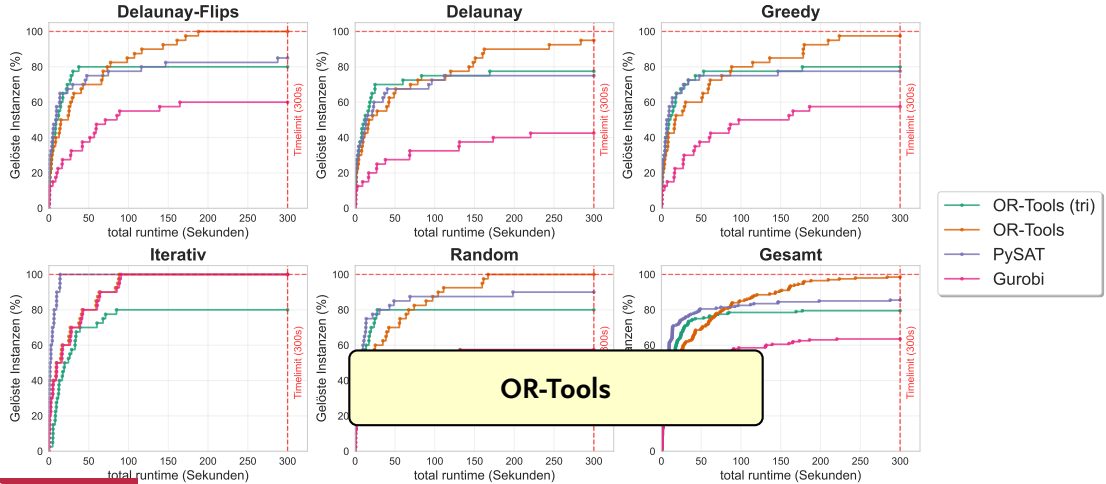


OR-Tools mit Kanten, Hülle Fixieren und alternative Kantenbedingung

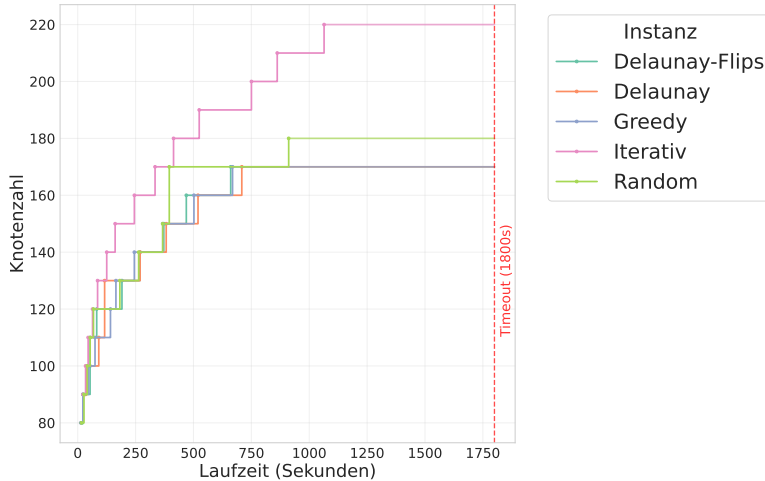
Solver Vergleich



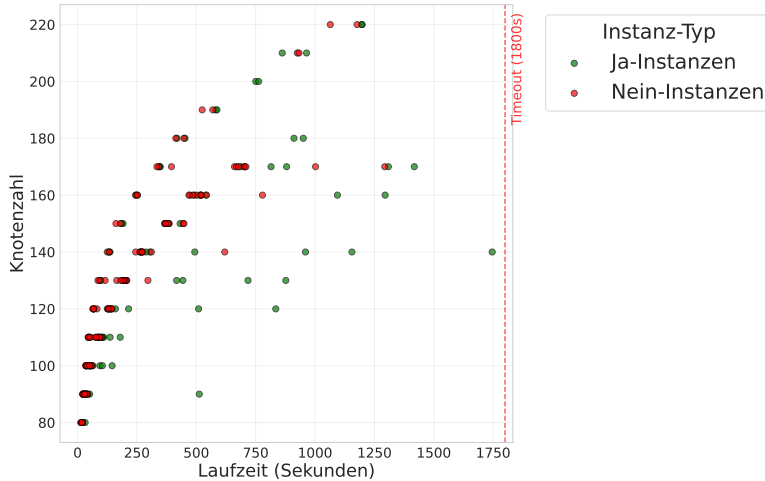
Solver Vergleich



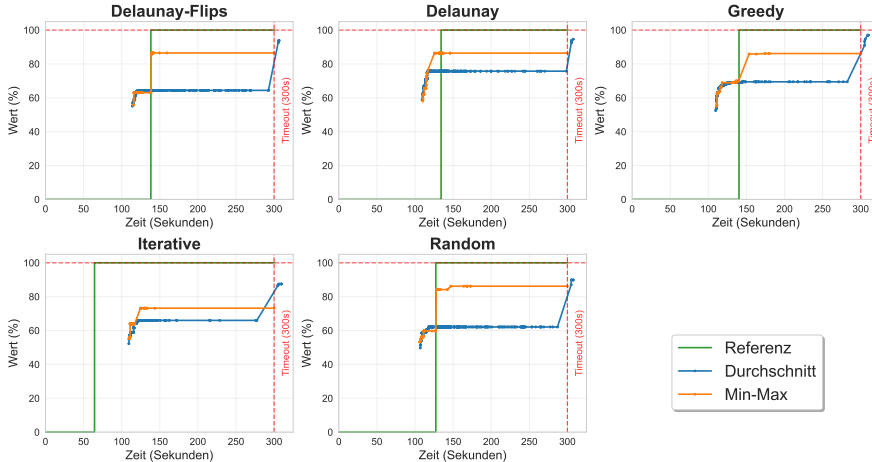
Größerer Timeout



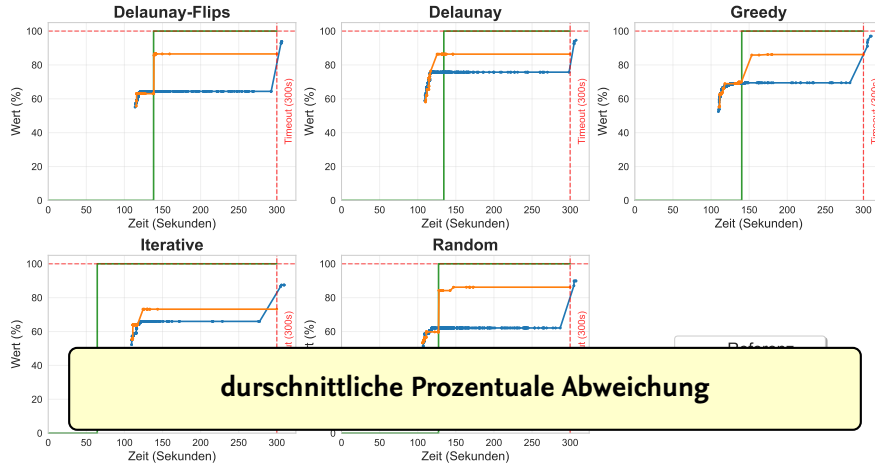
Ja Nein getrennt



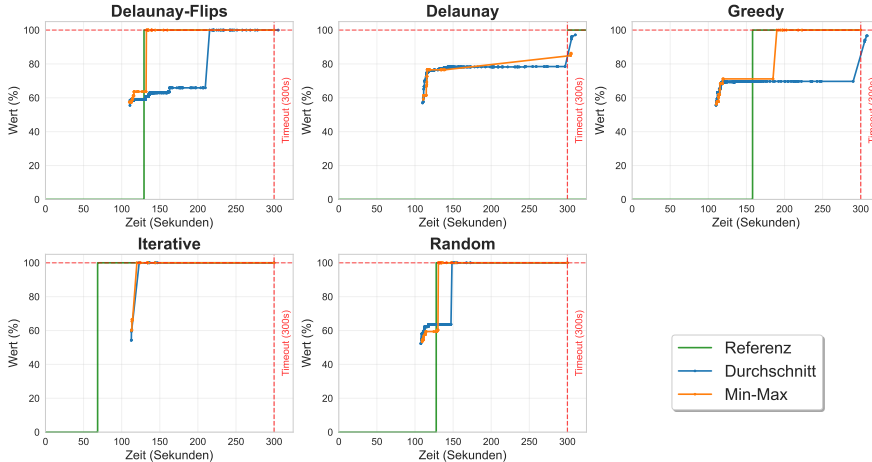
Nein Instanz



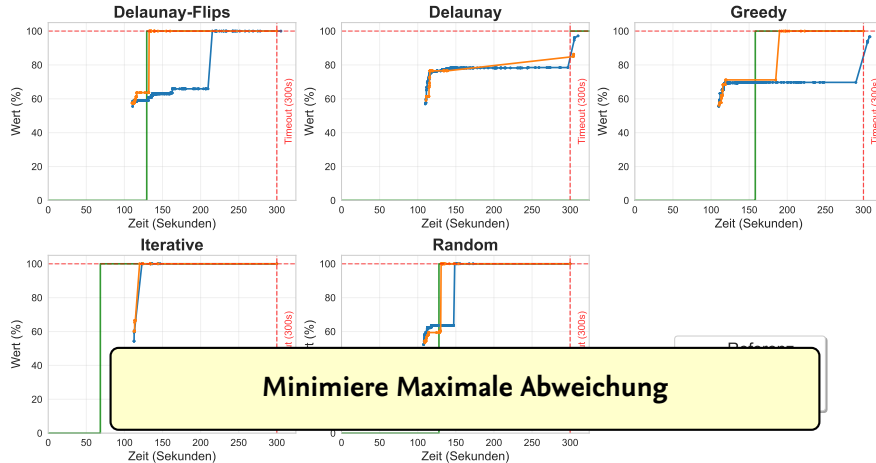
Nein Instanz



Ja Instanz



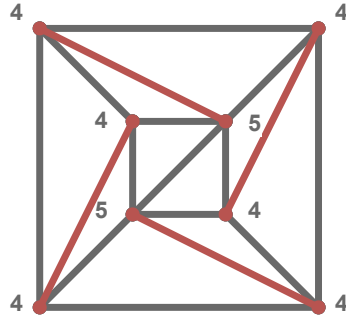
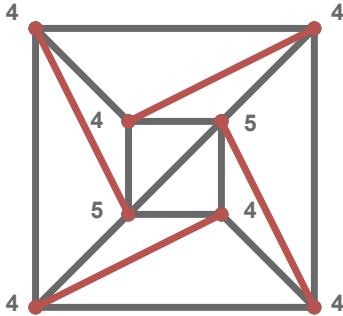
Ja Instanz



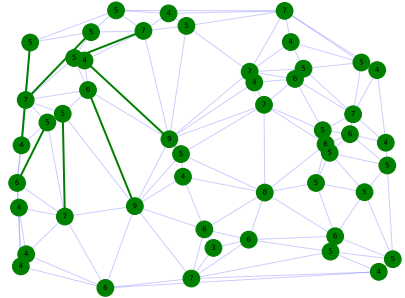
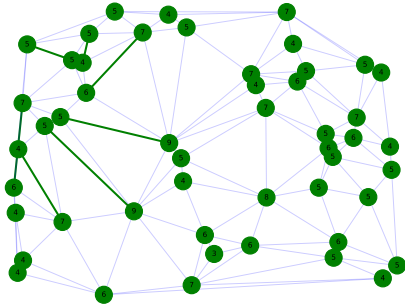
Einordnung

1. Iterativ
2. Random
3. Delaunay-Flips
4. Greedy
5. Delaunay

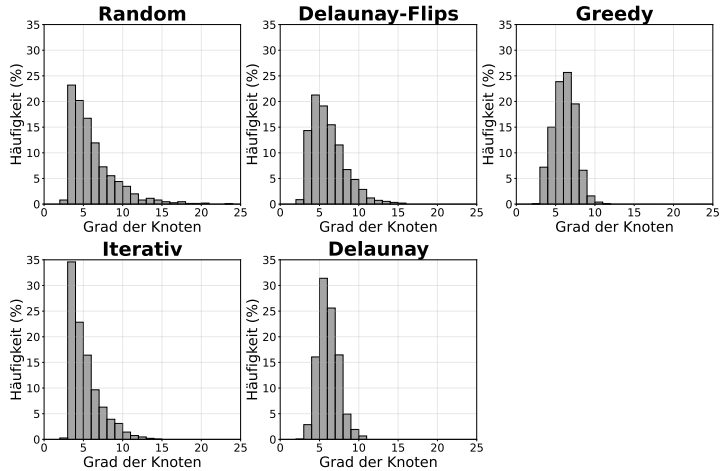
Anzahl Lösungen



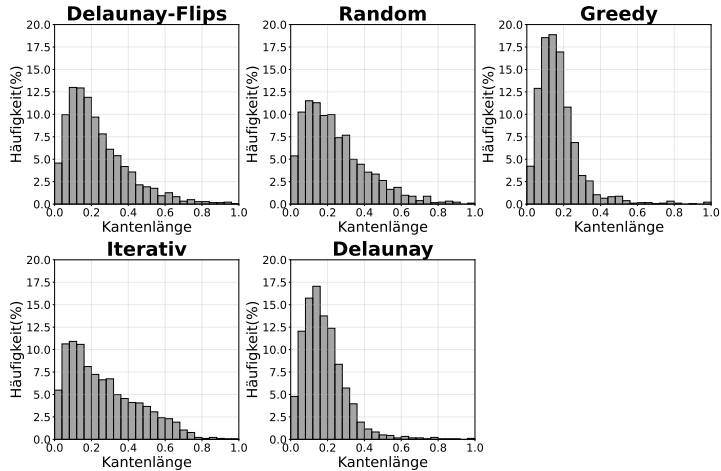
Anzahl Lösungen



Gradverteilung



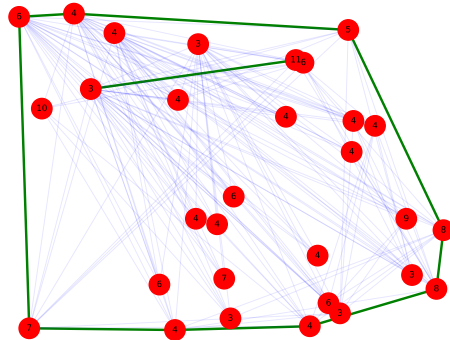
Kantenlängen



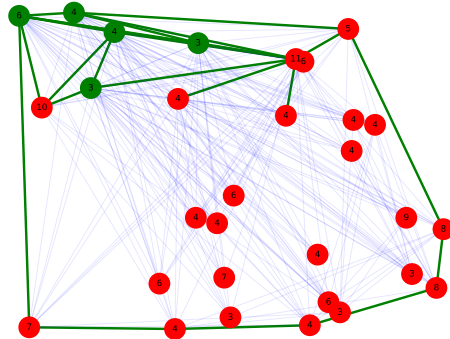
Conflict-driven clause learning

CaDiCaL Simplified Satisfiability Solver

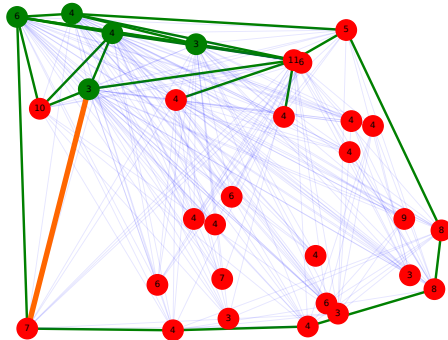
Teilinstanz



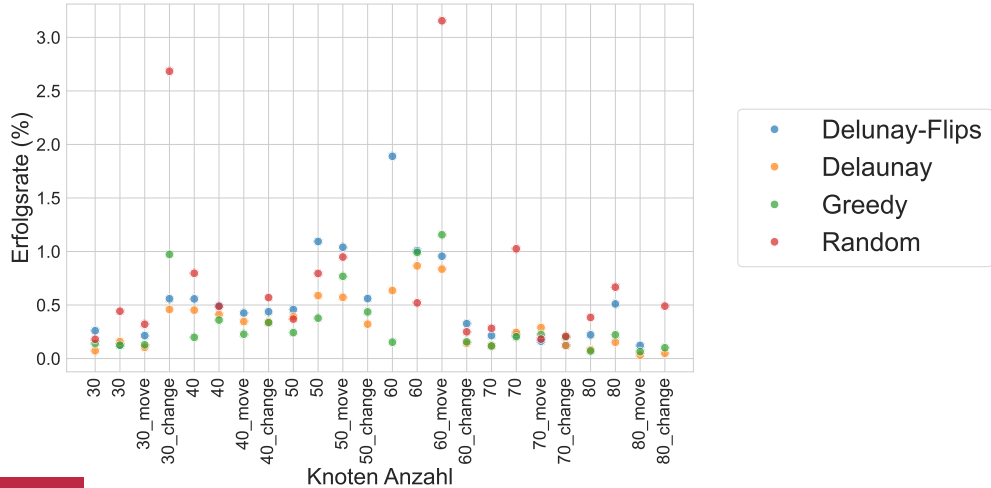
Teilinstanz



Teilinstanz



Propagation



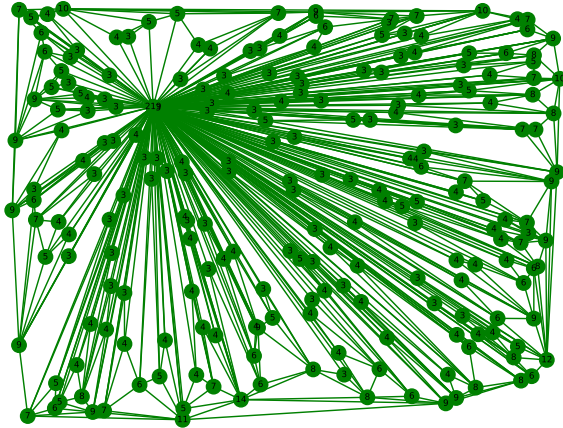
Teil IV

Fazit und Ausblick

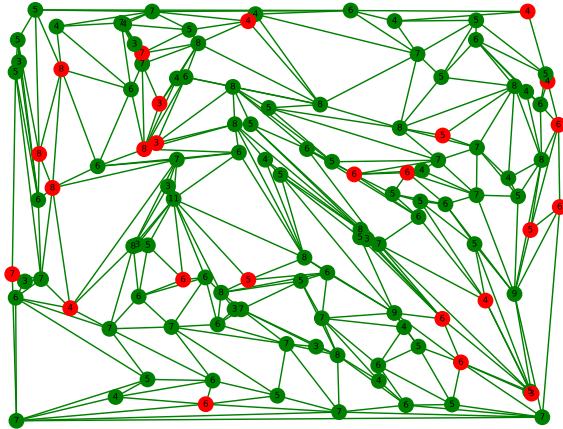
Fazit

OR-Tools Hülle Fixieren Kanten Fixieren alternative Kantenbedingung

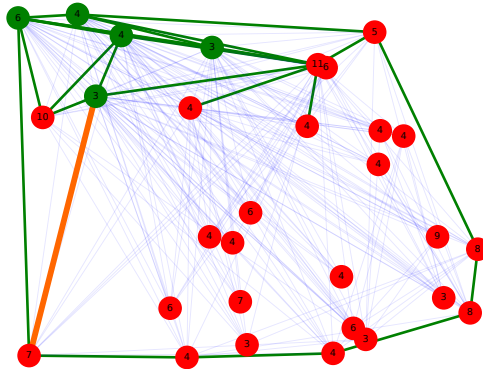
Fazit



Fazit



Fazit



Ausblick

- Skalierbarkeit
- Arbeitsspeicherverbrauch
- Dreiecksansatz Schwierigkeit Instanzen
- Optimierungen Kombinationen
- Propagation erweitern
- normalverteilte Triangulierung
- Problem abwandeln
 - Kanten vorgeben
 - Polygone